

IPSec Security Protocol

Module 14

Duration: 26:49

Video: https://www.youtube.com/watch?v=wVX5KLPh_o8

Course Coordinator - Dr Maninder Singh, Professor and Dean, Academic Affairs, Thapar University, Patiala

Subject Expert - Prof Yogesh Chaba, Department of Computer Engineering and IT, Guru Jambheshwar University of Science & Technology, Hisar

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0:00 [Music]

0:19 Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar

0:28 University of Science and technology Hisar harana today in this lecture we will

0:35 study about IP SEC security protocol first of all please see the contents

0:42 which I shall cover after covering brief introduction about IPC Internet Security

0:50 protocols I shall take up its architecture which will be followed by

0:56 IPC protocols ah that is authentication header and ESP that is

1:05 encapsulating security payload in last I shall cover ik that is internet key

1:14 exchange protocol for Key Management IP SEC Internet Protocol

1:23 security is a framework for a set of protocols for security at the network

1:30 or packet processing layer of network communication earlier security

1:38 approaches have inserted security at the application layer of the communication

1:45 model IPC is set to be especially useful for implementing virtual private networks

1:54 and for remote user access through dialup connection to PR private networks

2:01 2 minutes, 1 second a big advantage of IPC is that security Arrangements can be

2:09 handled without

2:09 requiring changes to individual user computers IPC is a protocol suit for

2:19 securing Internet Protocol IP communication by authenticating and encrypting

2:29 each IP

2:29 packet of a communication session IPC can be used in protecting

2:37 data flows between a pair of hosts means host to host between a pair of security

2:46 gateways means Network to network or between a pair of security Gateway and a

2:54 host means Network to host IP SEC is is not a single protocol

3:01 3 minutes, 1 second but rather a set of services and protocols that provide a

3:09 complete

3:09 Security solution for an IP network these services and protocols combine to

3:17 provide various types of protection some of the kinds of protection services
3:25 offered by IPC include the following it offers
3:32 confidentiality by encrypting data it provides Integrity by calculating checkm
3:40 or hash value of the data and authentication by using signatures and
3:50 certificates it provides protection against certain types of security
3:56 attacks such as replay attack it also manages sessions along with key
4:06 management now after covering basic introduction of Ip sec let us move to
4:13 its architecture RFC
4:16 [Music]
4:18 241 defines the basic architecture of Ip
4:23 SEC it covers Concepts in which security Association
4:30 sa a provide the bundle of algorithms and data that provide the parameters
4:39 necessary for operations of two IP protocols ah authentication header and
4:49 ESP encapsulating security payload there are two modes in which
4:57 these protocols work transport mode and tunnel mode in the
5:04 end there is Key Management protocol like internet key exchange that is ik
5:14 ke now let us first move on to modes of protocol in IP secc IPC can be
5:23 implemented in a host to host transport mode as well as in a Network tunneling
5:32 mode in transport mode only the payload of the IP packet is usually encrypted
5:40 and authenticated the routing is intact because the IP header is neither
5:48 modified nor encrypted in tunnel mode the entire IP
5:55 packet is encrypted or authenticated it is then encapsulated into a new IP
6:04 packet with a new IP header other way to explain difference
6:11 between transport mode and tunnel mode is that transport mode can only be used
6:19 between end points of a communication or we can say transport mode is used when
6:28 the cryptographic end points are also the communication end points of the
6:36 secured IP packets figure shows transport mode in
6:43 which communication and cryptographic end points are A and B tunnel mode can
6:53 be used between arbitrary peers when at least one cryptography traic Endo is not
7:01 7 minutes, 1 second a communication Endo of the secured IP
7:07 packets sa a indicates here security associations lower part of figure shows
7:17 tunnel mode in which communication takes place between either end point a and
7:25 router R A or router r a a to router
7:31 RB or router RB to endpoint B
7:39 accordingly packet structure is shown in which source is router R A and
7:46 destination is router RB tunnel mode is used to create virtual
7:54 packet networks for Network to network communications host to network
communications and host

8:04 to host Communications IPC provides two choices of security service authentication
8:13 header ah which essentially allows authentication of the sender of data and
8:21 encapsulating security payload ESP which supports both authentication
8:28 of the sender and encryption of data as well the specific information associated
8:37 with each of these Services is inserted into the packet in a header that follows
8:45 the IP packet header now let us take each one by one authentication header ah
8:55 is a member of the IPC protocol suit ah guarantees connectionless integrity and
9:04 data origin authentication of Ip packets further it can optionally protect
9:11 against replay attacks by using the sliding window technique and discarding
9:19 old packets ah is one of the two core security protocols in IP
9:27 SEC it provides authentication of either all or part of the contents of a
9:34 datagram through the addition of a header that is calculated based on the
9:41 values in the datag gr authentication header ah protocol work in both
9:49 transport and tunneling mode figure shows packet format
9:56 alteration with ah for Trans Port mode in this ah header is added after payload
10:05 that consist of TCP header and data next is figure which shows packet
10:13 format alteration with ah for tunnel mode in this ah header is added after
10:22 original Ip header and data here data
10:28 also contains TCP or UDP header ah uses a simple algorithm which
10:38 uses special hashing algorithm and a specific key known only to the source
10:46 and the destination security Association sa between two devices specifies these
10:56 particulars so that the source and destination know how to perform the
11:03 computation but nobody else can on the source device ah performs the
11:13 computation and puts result Integrity check value or
11:20 icv into a special header with other fields for transmission the destination device
11:29 does
11:36 the same calculation using the key that the two devices
11:43 share this enables the device to see immediately if any of the fields in the
11:53 original datagram were modified or not now let us move over to the format of
12:01 authentication header please C figure which shows different fields of of
12:10 authentication
12:18 12 minutes, 1 second header let us discuss each field one by one first field is next
12:27 header it is of
12:34 one byte which contains the protocol number of the next header after the
a it is used to link headers together next field is payload length this field
is of 1 byte and it measures the length of the authentication header itself
including the payload the length of this authentication header is measured in two

12:43 octet units minus 2 for example an ah
12:49 value of 4 equal $3 + 3 - 2$ here first
12:56 three indicates first three rows of 32 bit each and next three indicates three
13:05 rows for 32bit icv field and thus an ah
13:12 value of 4 means 24 octets Third Field is reserve which is
13:20 of 16 bits and is not used till now so it is set to zero next field is
13:29 security parameter index SPI which is of 32 bits and it is
13:37 arbitrary value which is used together with the destination IP address to
13:45 identify the security Association of the receiving
13:51 party security parameter index SPI is followed by sequence number field
13:58 which which is of 32 bits the field is incremented by one for
14:05 every packet sent to prevent replay attacks when replay detection is enabled
14:14 sequence numbers are never reused because a new security Association must
14:21 be renegotiated before an attempt to increment the sequence number Beyond its
14:30 maximum value last field is integrity check value
14:38 icv which is multiple of 32 bits this field may contain padding to align the
14:47 field to an 8 octet boundary for IPv6 or
14:53 a four octet boundary for ipv4
14:59 the IPC authentication header ah provides Integrity authentication
15:06 services to IP SEC capable devices so that they can verify that messages are
15:15 received intact from other devices for many applications however
15:23 this is only one piece of the puzzle we want to not only protect
15:29 against intermediate devices changing the datagram but also to protect against
15:37 them examining their contents as well for this level of private communication
15:46 ah is not enough we need to use the ESP protocol because ah just provides
15:56 authentication but not privacy privacy is provided by
16:04 es now let us see next protocol in IPC that is
16:11 encapsulating security payload that is es encapsulating security payload ESP is
16:21 a member of the IPC protocol suit IPC provides origin
16:29 authenticity Integrity confidentiality and protection of
16:36 packets ESP operates directly on top of Ip the main job of ESP is to provide the
16:46 Privacy we seek for IP datagram by encrypting them an encryption algorithm
16:55 combines the data in the datagram with a key to transform it into an encrypted
17:03 form ESP can also be implemented in both transport and tunneling
17:12 mode figure shows packet format alteration with ESP for transport mode
17:21 in this ESP header is added after payload that consist of TCP header and
17:28 and data ESP trailer and ESP authentication fields are added before

17:36 TCP header and data however in tunneling mode where the
17:43 entire original IP packet is encapsulated with a new packet header
17:51 added ESP protection is afforded to the whole inner IP packet while the outer
18:00 header remains unprotected as in transport mode ESP
18:06 trailers and ESP authentication are added before
18:13 data now let us move over to the format of encapsulating security payload ESP
18:22 please see the figure which shows different fields of
18:27 ESP let us discuss each field One By One ESP has several fields that are same
18:37 as those used in ah but it packages its field in a very different way instead of
18:47 having just a header it divides its Fields into three
18:54 components first is ESP header this contains two Fields SPI and sequence
19:03 number and they come before the encrypted data its placement depend on
19:11 whether ESP is used in transport mode or tunnel mode as explained earlier in my
19:20 lecture next is ESP trailer this section is placed after the encrypted data it
19:28 contains padding that is used to align the encrypted data through a padding and
19:36 Pad length field interestingly it also contains the next header field for
19:45 ESP ESP authentication data this field contains an
19:52 icv that is computed in such a manner that is similar to
19:59 how the ah protocol works the field is used when esp's optional authentication
20:06 feature is employed now let us see each of these
20:12 field one by one first field is security parameter index which is of 32 bits it
20:22 stores an arbitrary value used together with a destination IP address to
20:30 identify the security Association of the receiving party next field is sequence
20:39 number
20:47 which is also of 32 bits it is monotonically increasing sequence number
20:55 which increments by one for every packet sent to protect against replay
21:02 attacks there is a separate counter kept for every security
21:09 Association Next we move on to payload data field which is variable after
21:19 payload data it is padding which is of 0 to 255 octets it is used as padding for
21:26 encryption to extend the payload data to a size that fits the encryption Cipher
21:35 block size and to align the next field next field is pad length field which is
21:41 of 8 Bits And it specifies size of padding in
21:48 octets moving on to next header field which is of 8 Bits And contains next
21:57 Header information now there are two reasons why these fields are broken into
22:05 pieces
like this the first is that some encryption algorithms require the data to be
encrypted to have a certain block size and so padding must appear after the

22:12 data and not before it that's why padding appears in the ESP trailer field
22:21 the second is that the ESP authentication data appears separately
22:27 because it is used to authenticate the rest of the encrypted datagram after
22:35 encryption this means that it cannot appear in the ESP header or ESP trailer
22:43 next we shall discuss Key Management protocol in IP SEC which is internet key
22:51 exchange I ke like many secure networking protocol sets it is based on the concept
of a
23:01 23 minutes, 1 second shared secret two devices that want to send information
securely and code and
23:09 decode it using a piece of information that only the devices know anyone who
23:18 isn't on the secret is able to intercept the information but is prevented either
23:27 from reading it or from tempering the primary support protocol used for this
23:34 purpose in IPC is known as Internet key exchange that is ik
23:44 ke ik is defined in RFC 249 and it is more complicated of the
23:53 IPC protocols to comprehend in fact it is simply impossible to truly understand
more than
24:03 a real simplification of its operation without significant background in
24:11 cryptography the purpose of ik e is to allow devices to exchange
24:18 information that is required for secure communication as the title suggest this
24:27 includes cryptographic keys that are used for encoding authentication information
and
24:36 Performing payload encryption ik works by allowing IPC capable devices to
24:45 exchange SAS main feature of ik are as shown in
24:54 figure ik is an IPC component used for performing Mutual authentication and
25:03 establishing and maintaining security associations it is typically used for
25:10 establishing ipx sessions it has a key exchange mechanism for security at
25:18 Network layer it has five variations of an ik negotiation with two
25:27 modes aggressive and Main it also has three authentication
25:35 methods pre-shared public key encryption and public key
25:43 signature so dear friends today in this lecture we have covered introductory
25:49 concepts of IPC protocol followed by its architecture then we covered IP Che
25:58 protocols ah that is authentication header and ESP that is
26:05 encapsulating security payload in the last ik that is internet key exchange
26:13 protocol for Key Management was covered thank you
26:31 [Music]
26:42 [Music]